

Active Maintenance Report: peyton_huber_coppock_2022

Peyton, Huber and Coppock (2022, JEPS): The Generalizability of Online Experiments Conducted During the COVID-19 Pandemic

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This repository holds the actively maintained replication code for Peyton, Huber and Coppock (2022), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi [10.1017/S1049096521000366](https://doi.org/10.1017/S1049096521000366)) to a set of published archives.

Article	10.1017/xps.2021.17
Corrigendum	10.1017/xps.2022.23
Replication archive	10.7910/DVN/38UTBF

The data are not redistributed here. The deposit is 93 MB across 20 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published figure or table, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This README is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See LICENSE.

To reproduce. Clone or download the repository, open `peyton_huber_coppock_2022.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 20 files, produces every figure and table into `maintained/output/`, runs the deposited code once more to record what it produces, rebuilds the ground truth and runs the coverage gate over the in-text claims. Required packages: tidyverse, estimatr, metafor, ggthemes, ggrepel, ggforce, lemon, RCurl, haven, xtable, knitr, kableExtra, here. Paths resolve through **here**, so nothing depends on the working directory. The run also needs network access beyond Dataverse, because the deposited

appendix code downloads two ManyLabs 2 datasets from GitHub. A successful run overwrites `maintained/output/`, which is committed: **git diff on that folder is the reproduction check.**

1 Paper overview

Citation: Peyton, K., Huber, G. A. and Coppock, A. (2022). “The Generalizability of Online Experiments Conducted During the COVID-19 Pandemic.” *Journal of Experimental Political Science*, 9(3), 379-394. DOI: 10.1017/xps.2021.17

Corrigendum: Peyton, K., Huber, G. A. and Coppock, A. (2023). *Journal of Experimental Political Science*, 10(2), 306-309. DOI: 10.1017/xps.2022.23

Summary: The paper asks whether survey experiments fielded during the COVID-19 pandemic recover what the same designs found before it. The authors fielded 33 replications of 12 previously published designs across 13 weekly Lucid quota samples of roughly 1,000 US respondents each, between March and July 2020, spanning binary, ordinal, six-arm and factorial conjoint designs. Pre-COVID benchmarks come from the original papers, from prior replications including ManyLabs 2, and from archived data; Glass’s Delta standardizes outcomes within study, and summary effect sizes pool treatment-outcome pairs by inverse-variance weighting. Effects replicate in sign and significance but at reduced magnitude: replication estimates average 73 percent of their pre-COVID benchmarks, 87 percent among conjoint designs and 49 percent among the rest. Inattention, proxied by attention-check failure, mobile devices and web-application respondents, accounts for much of the attenuation, since attentive subgroups recover effect sizes close to the pre-COVID benchmarks.

2 Summary

Two questions, answered before the detail.

2.1 Does the deposited archive run?

Not as deposited, but only one of the failures is more than an inconvenience.

The trivial ones are three. `ReadMe.R`, which is the entry point, loads `attentive`, a package that has only ever existed on GitHub, and `coefplot`, which is on CRAN. Neither package’s functions are called anywhere in the archive, so both `library()` calls can simply be dropped. `manuscript.R` reads `phc_summary.rds`, which the deposit does not contain because

`appendix_section_a.R` generates it, so the scripts have to be run in the order `ReadMe.R` names. And `manuscript.R` calls `scales::label_number_si()`, which `scales` made defunct in 2022 and which has a one-line replacement.

The fourth is substantive. `appendix_section_a.R` ends by writing the file every later script depends on, and the last statement before that write collapses a grouping variable with `paste()` where it should take one value per group. Under the dplyr of 2021 this silently produced a character vector of run-together study names; under dplyr 1.1 and later it is an error, and the file is never written. The bug is one word long and sits at the very end of a 3,010-line script, so nothing announces itself until the whole thing has run.

One further failure is worth separating out because it is a fact about a dependency rather than about the archive. `appendix_section_a.R` and `manuscript.R` call `position_dodgev()`, which the archive obtains through `lemon`. `lemon` is current on CRAN, updated in September 2025, but it no longer re-exports that function from `ggstance`. The archive is not broken; the package moved.

2.2 Does the maintained rewrite reproduce the paper?

Almost everywhere. The ground truth carries 426 rows; 329 of the 358 the rewrite can compute agree with the published value at the page’s own precision, and 53 of the 59 claims about shape, sign or count hold. That includes every count the 2023 corrigendum corrected. The 35 that do not are among the rows the number-by-number comparison below lists, and eleven of them are corrected in `peyton_huber_coppock_2022_errata.pdf`.

Two things are worth stating. First, the deposited archive is **not** the version that produced the erroneous counts in the published article. Harvard Dataverse version 2.0, released 6 October 2022, replaced `manuscript.R` with the corrected code, and running it reproduces the corrigendum’s numbers exactly. Second, `lemon` and `ggforce` are both current CRAN packages, so nothing in this archive depends on an abandoned graphics package.

The deposit does not reproduce one of its own published tables. `appendix_section_a.R` pools the COVID-era weeks of the prospective atomic aversion experiment without restricting them to the unstandardized estimates, so each week enters the precision-weighted mean twice, once on the percentage-point scale and once on Glass’s Delta scale. 17 cells of the appendix’s Table 2 move as a result. The three blocks around it are written the same way with the filter present, and restoring it returns all 44 of that table’s cells exactly.

The one genuinely stale dependency is `rmeta`, whose most recent release is from March 2018. The maintained rewrite replaces it with `metafor`, which changes no result: see the section on fixed-effect and random-effects pooling.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 in August 2026.

Script	Status on current R	Resolution
ReadMe.R	Fails at library(attentive)	Drop library(attentive) and library(coefplot); neither package's functions are called anywhere in the archive
helpers.R	Clean (sourced by all scripts)	No changes required
appendix_section_a.R	Fails twice	position_dodgev() is no longer re-exported by lemon, so supply the position_dodge equivalent; then paste(study_group) must become first(study_group) to write phc_summary.rds
appendix_section_b.R	Clean	No changes required; deprecation warnings only
manuscript.R	Fails twice	Run appendix_section_a.R first so phc_summary.rds exists; replace the defunct scales::label_number_si() with label_number(scale_cut = cut_short_scale())

Everything else in the archive is a deprecation warning rather than a failure: `size` where `ggplot2` now wants `linewidth`, `geom_errorbarh()` superseded by an oriented `geom_errorbar()`, `do()` and `gather()` and `spread()` superseded within `dplyr` and `tidyr`, and `lemon::facet_rep_wrap()` soft-deprecated in favour of `ggh4x::facet_wrap2()`. All of them still run.

The `paste(study_group)` bug deserves emphasis because of what it did rather than what it does. In `dplyr` 1.0 it did not error. It returned a character vector, and `summarise()` recycled it, so `phc_summary.rds` was written with a `study_group` column whose entries were several study names run together. Nothing downstream reads that column in a way that would notice. The bug became visible only when a later `dplyr` turned the silent recycling into an error, which is to say the archive was more likely to be caught by a version change than by anyone reading its output.

4 Errata

4.1 Corrected p-value adjustment (published article, Figures 2 and 3 and page 6)

The published article reported the wrong number of statistically significant differences between the pre-COVID and COVID-era summary estimates. Errors in the code that adjusted p-values for multiple comparisons produced counts that were inconsistent before and after adjustment. The corrigendum (Peyton, Huber and Coppock 2023, *JEPS* 10(2), 306-309, doi 10.1017/xps.2022.23) corrects the page 6 text and both figures. No point estimate, standard error or substantive conclusion changes.

Table 2: Significant-difference counts corrected by the 2022 corrigendum. The deposited archive already carries the corrected code.

Quantity	Published article	Corrigendum	Deposited archive	Maintained rewrite
Non-conjoint: significantly smaller, of 24 correctly signed	10	11	11	11
Non-conjoint: significant at $p < 0.05$ (Figure 2 caption)	13	14	14	14
Non-conjoint: still significant after BH-FDR (Figure 2 caption)	not reported	13	13	13
Conjoint: significantly smaller, of 35 smaller	6	11	11	11
Conjoint: significant at $p < 0.05$ (Figure 3 caption)	7	12	12	12
Conjoint: still significant after BH-FDR (Figure 3 caption)	not reported	7	7	7

The deposit was updated alongside the corrigendum. Harvard Dataverse version 2.0, released 6 October 2022, replaced `ReadMe.R` and `manuscript.R`, and the replacement `ReadMe.R` says so in a comment. Running the deposited `manuscript.R` reproduces the corrigendum’s counts exactly, and so does the maintained rewrite, which computes them in `maintained/text_correspondence_summary.R` and writes them to `output/text_correspondence_summary.csv`. Reproducing the published article’s original counts is not possible from the deposit, because the code that produced them is no longer in it.

4.2 Everything else the reanalysis found

Eleven further corrections are set out in `peyton_huber_coppock_2022_errata.pdf` at the root of this repository, each quoting the published sentence and the corrected one, with every corrected value computed at render time from `maintained/output/`. None of them changes a conclusion. The largest is a paragraph of the online appendix that attributes four estimates to the original atomic aversion study which are in fact the pre-COVID replication’s, as the appendix’s own Table 2 shows.

4.3 A rounding difference in Table 2

One cell of published Table 2 differs from the pass rate behind it in its last digit. The Easy attention-check pass rate among web-application respondents is 0.5449. Rounded once to two decimal places that is 0.54, which is what the maintained rewrite prints. The deposited script rounds every estimate to three decimals before formatting it to two, so 0.5449 becomes 0.545 and then 0.55, which is what the published table shows. The deposit therefore reproduces its own table, and the discrepancy is between the printed cell and the quantity it reports. Nothing else in the table or the text depends on it. It is a table cell rather than a sentence, so it stays here rather than in the errata.

4.4 Two findings that are not corrections

Two disagreements are recorded in the ground truth without a correction, because establishing what the published figure should be would mean choosing between analytical conventions the paper does not state.

The appendix reports three pairwise comparisons between two of the authors' own replication arms, in Sections A.2 and A.7, and each printed p-value is the one-tailed value at the standard error the comparison implies: 0.02 against a two-sided 0.044, 0.09 against 0.183, and 0.46 against 0.919. Every other p-value in the article and the appendix is two-sided, and the deposited code computes two-sided p-values everywhere. The estimates and standard errors in those sentences are unaffected.

The annual device shares on page 10 reproduce for 2018 and 2020 and not for 2019. Averaging the per-survey shares in the deposited `phc_meta_trends.rds` over the surveys fielded in each year returns 13 per cent and 33 per cent for 2018 and 41 per cent and 56 per cent for 2020, exactly as printed, and 34 per cent and 58 per cent for 2019, against a printed 33 and 56. The 56 the sentence gives for 2019 is the same figure it gives for 2020 in the preceding clause.

5 The extraction and the two instruments

`ground_truth/published_claims.csv` is an exhaustive numeric-token extraction from the article, its 35-float online appendix and the 2023 corrigendum, read line by line rather than searched for expected patterns. It carries 440 rows, each classified as a **pipeline**, **descriptive**, **definitional**, **structural** or **transcribed** claim and each carrying the precision at which the page prints it, so that the two instruments cannot disagree about what precision a comparison should use. 276 of those rows require a recomputation.

The coverage boundary is every number printed in the body of the article, in every caption, note and footnote, and in the online appendix's prose and its three tables, plus every number in the corrigendum. Five classes of token are excluded and named here so the exclusion is auditable: bibliographic tokens, which is to say citation years and the volume, issue and page numbers of the two reference lists; page numbers and the page references in the appendix's contents; the DOIs and the institutional review board protocol number; axis tick labels, which are scale furniture rather than estimates; and the prices, probabilities and casualty counts quoted from the original studies' instruments, which the extraction records once per section as a **transcribed** row rather than one row per token.

`maintained/in_text_claims.R` is the second instrument. It recomputes every one of those 276 claims from `maintained/output/` by a route of its own and prints one line per claim. Where `build_ground_truth.R` reaches a quantity through a summary file, the claims file goes

back to the study-level estimates the summary was built from, so the two derivations are separate; where they disagree, one of them is wrong. `build_ground_truth.R` runs the claims file inside its own environment, captures what it prints, and halts unless the printed identifiers are exactly the identifiers the extraction requires and every printed value equals the ground truth's at the extraction's precision. It also halts if a transcribed value does not survive a round trip through its own recorded precision, if a published cell has no rewrite counterpart or a rewrite cell no published counterpart, if a float prints a different number of cells from the count the extraction declares, or if an adverse row carries no cause or a clean match carries one.

6 Number-by-number comparison

The ground truth carries 426 rows: 260 prose and caption quantities named in the extraction, 146 cells transcribed from Table 2 and the appendix's three tables, four relative sizes the appendix prints as a dash, and 16 counts of what a wordless figure plots. The full table is `ground_truth/peyton_huber_coppock_2022_ground_truth.csv`; only the rows that disagree are printed here.

Table 3: Every row of the ground truth that does not agree with the published page: 52 of 426.

Location	Quantity	Paper	Rewrite	Cause
Design (p. 4)	design_median_duration	12.8	12.88	paper_internal
Results (p. 5)	results_replication_estimates	101	102	unresolved
Results (p. 6)	results_sig_smaller_of_24	10	11	paper_internal
Results (p. 6)	results_conjoint_sig_among_smaller	6	11	paper_internal
Results (p. 6)	results_correspondence_nonconjoint	49	48.3	paper_internal
Figure 2 note (p. 7)	figure_2_sig_count	13	14	paper_internal
Figure 3 note (p. 8)	figure_3_sig_count	7	12	paper_internal
Inattention (p. 10)	device_2019_webapp	33	33.82	paper_internal
Inattention (p. 10)	device_2019_mobile	56	57.62	paper_internal
Inattention (p. 10)	device_overlap_webapp_given_mobile	72	74.51	unresolved
Inattention (p. 10)	device_duration_browser	21.5	21	paper_internal
Inattention (p. 11)	inattention_inattentive_share	26	27	paper_internal
Online Appendix A.1 (p. 1)	a1_replication_estimate	25.5	25.39	paper_internal
Online Appendix A.1 (p. 1)	a1_difference	5.5	5.108	paper_internal
Online Appendix A.2 (p. 3)	a2_covid_vs_precovid_p	0.02	0.04349	paper_internal
Online Appendix A.2 (p. 3)	a2_covid_vs_direct_p	0.09	0.183	paper_internal
Online Appendix A.7 (p. 15)	a7_covid_vs_direct_se	0.04	0.05695	paper_internal
Online Appendix A.7 (p. 15)	a7_covid_vs_direct_p	0.46	0.9177	paper_internal
Online Appendix A.4 (p. 6)	a4_capi_summary	1.00	0.9943	paper_internal
Online Appendix A.8 (p. 17)	a8_9070_success	75	70	paper_internal
Online Appendix A.8 (p. 18)	a8_original_9070_prefer	37	29.71	paper_internal
Online Appendix A.8 (p. 18)	a8_original_9045_prefer	51	47.56	paper_internal
Online Appendix A.8 (p. 18)	a8_original_9070_approve	17	5.136	paper_internal
Online Appendix A.8 (p. 18)	a8_original_9045_approve	27	27.52	paper_internal
Online Appendix A.9 (p. 22)	a9_total_amces	81	82	paper_internal
Online Appendix A.10 footnote 5 (p. 25)	a10_footnote_5_significant	2	4	paper_internal

Table 3: Every row of the ground truth that does not agree with the published page: 52 of 426.
(continued)

Location	Quantity	Paper	Rewrite	Cause
Corrigendum Figure 1a note (p. 307)	corrigendum_figure_1a_note_sig	13	14	paper_internal
Corrigendum Figure 2a note (p. 308)	corrigendum_figure_2a_note_sig	7	12	paper_internal
Results (p. 6)	results_all_smaller_nonconjoint	NA	FALSE	paper_internal
Inattention (p. 10)	device_additional_sample_period	NA	FALSE	paper_internal
Inattention (p. 10)	device_acq_significance	NA	FALSE	paper_internal
Online Appendix A.5 (p. 11)	a5_table_reference	NA	FALSE	paper_internal
Online Appendix A.5 (p. 10)	a5_program_a_significant	NA	FALSE	unresolved
Online Appendix A.10 footnote 5 (p. 25)	a10_footnote_5_named	NA	FALSE	paper_internal
Table 2	table_2_easy_webapp_estimate	0.55	0.5449	archive
Appendix Table 2	appendix_table_2_90_70_prefer_nuclear_use_ycls_summary_estimate	0.08	0.07875	archive
Appendix Table 2	appendix_table_2_90_70_prefer_nuclear_use_psv_diff_estimate	-0.22	-0.2184	archive
Appendix Table 2	appendix_table_2_90_70_prefer_nuclear_use_abp_diff_estimate	-0.29	-0.2889	archive
Appendix Table 2	appendix_table_2_90_70_prefer_nuclear_use_abp_diff_se	0.04	0.03582	archive
Appendix Table 2	appendix_table_2_90_45_prefer_nuclear_use_ycls_summary_estimate	0.13	0.1266	archive
Appendix Table 2	appendix_table_2_90_45_prefer_nuclear_use_ycls_summary_se	0.03	0.02599	archive
Appendix Table 2	appendix_table_2_90_45_prefer_nuclear_use_psv_diff_estimate	-0.35	-0.349	archive
Appendix Table 2	appendix_table_2_90_45_prefer_nuclear_use_psv_diff_se	0.06	0.05519	archive
Appendix Table 2	appendix_table_2_90_45_prefer_nuclear_use_abp_diff_estimate	-0.39	-0.3883	archive
Appendix Table 2	appendix_table_2_90_70_approve_nuclear_use_ycls_summary_estimate	-0.04	-0.03862	archive
Appendix Table 2	appendix_table_2_90_70_approve_nuclear_use_psv_diff_estimate	-0.09	-0.08998	archive
Appendix Table 2	appendix_table_2_90_70_approve_nuclear_use_abp_diff_estimate	-0.21	-0.212	archive
Appendix Table 2	appendix_table_2_90_45_approve_nuclear_use_psv_diff_estimate	-0.32	-0.3203	archive
Appendix Table 2	appendix_table_2_90_70_prefer_nuclear_use_psv_ratio	0.27	0.2651	archive
Appendix Table 2	appendix_table_2_90_70_prefer_nuclear_use_abp_ratio	0.21	0.2142	archive
Appendix Table 2	appendix_table_2_90_45_prefer_nuclear_use_psv_ratio	0.27	0.2663	archive
Appendix Table 2	appendix_table_2_90_45_prefer_nuclear_use_abp_ratio	0.25	0.2459	archive

6.1 Float coverage

Table 4: Published floats that print a number or plot an estimate. A float printing no numbers is covered by the count of what it plots; the remaining floats, listed in the float coverage file, carry a stated reason.

Float	Numbers printed	Covered	Reproduced by the rewrite	Reproduced by the deposit	Estimates plotted
table_2	36	36	35	36	NA

Table 4: Published floats that print a number or plot an estimate. A float printing no numbers is covered by the count of what it plots; the remaining floats, listed in the float coverage file, carry a stated reason. (*continued*)

Float	Numbers printed	Covered	Reproduced by the rewrite	Reproduced by the deposit	Estimates plotted
figure_2	0	0	0	0	28
figure_3	0	0	0	0	41
figure_5	0	0	0	0	10
appendix_table_1	42	42	42	42	NA
appendix_table_2	44	44	44	27	NA
appendix_table_3	24	24	24	24	NA
figure_a1	0	0	0	0	4
figure_a2	0	0	0	0	4
figure_a3	0	0	0	0	10
figure_a4	0	0	0	0	34
figure_a5	0	0	0	0	24
figure_a6	0	0	0	0	2
figure_a7	0	0	0	0	4
figure_a8	0	0	0	0	20
figure_a9	0	0	0	0	10
figure_a10	0	0	0	0	82
figure_a11	0	0	0	0	12
figure_a12	0	0	0	0	10
figure_a13	0	0	0	0	8

The 24 floats not in that table are the eight covariate-distribution figures of appendix Section B, whose proportions the rewrite does not compute; the eleven Section C figures, which are screenshots of survey content and state no estimated quantity; Table 1 and Figures 1 and 4, which print no estimates and whose content is covered as prose claims; and the corrigendum’s two figures, which reprint the article’s.

7 Maintained rewrite

The rewrite (*maintained/*) is eleven scripts: a shared `helpers.R`, one cleaning script, five figures, one table, two scripts for quantities the article states only in prose, and the in-text claims file that recomputes every one of them.

Table 5: Maintained rewrite scripts and their outputs.

Script	Output
<code>helpers.R</code>	Packages, <code>theme_ycls()</code> , <code>meta_summaries_fe()</code> , <code>glass_delta()</code> , <code>make_entry()</code> , <code>table_entry()</code>

<code>clean_summary_estimates.R</code>	<code>phc_summary_clean.rds</code> and <code>.csv</code> ; <code>appendix_study_estimates.csv</code> ; <code>appendix_table_cells.csv</code> ; <code>appendix_table_ratios.csv</code> ; <code>text_pooled_benchmarks.csv</code>
<code>figure_1_lucid_weekly_completes.R</code>	<code>figure_1_lucid_weekly_completes.pdf</code> , <code>.png</code> and <code>.csv</code>
<code>figure_2_noncojoint_correspondence.R</code>	<code>figure_2_noncojoint_correspondence.pdf</code> , <code>.png</code> and <code>.csv</code>
<code>figure_3_conjoint_correspondence.R</code>	<code>figure_3_conjoint_correspondence.pdf</code> , <code>.png</code> and <code>.csv</code>
<code>figure_4_meta_trends.R</code>	<code>figure_4_meta_trends.pdf</code> , <code>.png</code> and <code>.csv</code>
<code>figure_5_trust_replication.R</code>	<code>figure_5_trust_replication.pdf</code> , <code>.png</code> and <code>.csv</code>
<code>table_2_acq_pass_rates.R</code>	<code>table_2_acq_pass_rates.tex</code> and <code>.csv</code> ; <code>table_2_acq_pass_rates_cells.csv</code>
<code>text_correspondence_summary.R</code>	<code>text_correspondence_summary.csv</code>
<code>text_device_metadata.R</code>	<code>text_device_metadata.csv</code> ; <code>text_device_by_survey.csv</code> ; <code>text_device_by_year.csv</code> ; <code>text_device_all_surveys.csv</code>
<code>in_text_claims.R</code>	Printed to the console: one line per published claim

7.1 Architecture

Every comparison in the paper rests on 138 summary effect sizes, one pre-COVID and one COVID-era estimate for each of 69 study-outcome pairs. The deposited `appendix_section_a.R` builds them in 3,010 lines of study-by-study estimation, and only its last statement is wrong. `clean_summary_estimates.R` therefore runs the deposited code up to that point and then applies the corrected summarise, rather than reimplementing twelve studies' worth of estimation to fix one word. That is a deliberate choice and it is the one place where the rewrite depends on the deposit as code rather than as data.

Running someone else's script inside a rewrite needs care, and three things are done to contain it. The deposited code addresses its inputs by bare relative path and writes appendix tables and figures beside them, so it runs in a temporary directory of symlinks to `original/`: reads resolve, writes land in `tempdir()`, and `original/` stays byte-identical to what `download_original.R` verified. The environment it evaluates in is passed explicitly rather than inherited. And three substitutions are applied to the text of the script itself. Two are dependency repairs, described in the next section. The third is an analysis correction: the block that pools the COVID-era weeks of the prospective atomic aversion experiment omits the `estimate_type == "Unstandardized"` filter that its three siblings in the same file all carry, so it averages each week's percentage-point estimate together with the same week's Glass's Delta estimate. Restoring the filter is what makes the rewrite reproduce the appendix's published Table 2.

Every figure script also writes a CSV of the estimates it plots. A PDF records the time it was written, so for a paper whose results are almost entirely figures the committed output would otherwise be undiffable.

The remaining scripts read either `original/` or the cleaned summary estimates, and share nothing but `helpers.R`. `ground_truth/extract_archive_values.R` runs the deposited code

a second time with only the repairs it needs to execute, and not the analysis correction, so that the deposit's own answer is recorded separately from the rewrite's.

7.2 Deprecated patterns replaced

Table 6: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
<code>'rmeta::meta.summaries()'</code>	<code>'metafor::rma(method = "FE")'</code>
<code>'position_dodgev()'</code> (ggstance, via lemon)	<code>'position_dodge(width = 0.5)'</code>
<code>'geom_errorbarh()'</code>	<code>'geom_linerange(xmin =, xmax =)'</code>
<code>'facet_row()'</code> (ggforce)	<code>'facet_wrap()'</code>
<code>'scales::label_number_si()'</code>	<code>'label_number(scale_cut = cut_short_scale())'</code>
<code>'do(tidy(...))'</code>	<code>'reframe(tidy(lm_robust(..., data = pick(everything()))))'</code>
<code>'paste(study_group)'</code> in <code>'summarise()'</code>	<code>'first(study_group)'</code>
<code>'size ='</code> for lines	<code>'linewidth ='</code>
<code>'ifelse()'</code>	<code>'if_else()'</code>
<code>'xtable()'</code> + <code>'print.xtable()'</code>	<code>'knitr::kable()'</code> + <code>'readr::write_lines()'</code>
<code>'%>%'</code>	<code>'&#124;>'</code>

Three of these deserve a word. `position_dodgev()` came from `ggstance`, which `ggplot2` superseded once `position_dodge()` learned to dodge along a discrete axis; `lemon` re-exported it when this archive was deposited and no longer does. `facet_row()` still exists in current `ggforce`, and the replacement is a simplification rather than a repair: with two panels `facet_wrap()` lays them out identically and the maintained figure needs one fewer package. `coord_capped_cart()`, also from `lemon`, is left in place in the deposited code the cleaning script runs, since `lemon` is maintained and the call affects only appendix figures drawn into a temporary directory.

The rewrite does not use `xtable`, but the deposited appendix code does, at five sites that write LaTeX table bodies with `only.contents = TRUE`. Those are fragments with no header, no row or column names and no rules, meant to be read into a hand-written `tabular` environment in the appendix. `knitr::kable()` has no equivalent output mode, so converting them would mean pasting the body rows together by hand rather than swapping one call for another. They are left as deposited, and `xtable` remains a dependency of `clean_summary_estimates.R` for that reason alone.

8 Fixed-effect and random-effects pooling

The deposited appendix code pools benchmark estimates with `rmeta::meta.summaries()` at fifteen sites, always at its default of fixed-effect inverse-variance weighting. `rmeta`'s most recent release is from March 2018, so its continued availability is a fact about CRAN's archiving policy rather than about maintenance. The rewrite replaces it with `metafor::rma(method = "FE")`, which is the same estimator: across all fifteen sites the largest disagreement in a pooled estimate is 7 times 10 to the minus 16, and in a standard error 7 times 10 to the minus 18, which is double-precision noise.

Having the data in `metafor` makes the random-effects estimates free, and they are reported below as an addition rather than a correction. The published article uses the fixed-effect column throughout, and those are the numbers that reproduce.

Table 7: Pooled benchmark estimates. FE is inverse-variance fixed-effect pooling, the estimator the article uses; RE is REML. $\hat{\tau}^2$ is the REML estimate of between-study variance and I^2 the share of total variance attributable to it. Q is Cochran's heterogeneity statistic, which the appendix quotes for the three welfare survey modes.

Benchmark	k	FE est.	FE SE	RE est.	RE SE	$\hat{\tau}^2$	I^2	Q	p
Russian reporters, pre-COVID	3	0.305	0.020	0.280	0.056	0.0074	83.3%	15.33	0.000
Question framing, pre-COVID	2	0.174	0.015	0.269	0.115	0.0240	90.7%	10.70	0.001
Question framing, COVID era	2	0.111	0.031	0.111	0.041	0.0015	43.6%	1.77	0.183
Asian disease, pre-COVID	5	0.289	0.014	0.325	0.043	0.0081	88.8%	24.97	0.000
Asian disease, COVID era	5	0.147	0.019	0.147	0.019	0.0000	0.0%	2.22	0.695
Welfare spending, COVID-era web interviews	12	0.437	0.017	0.437	0.019	0.0007	17.1%	12.72	0.312
Welfare spending, GSS in-person	9	0.994	0.012	0.993	0.018	0.0016	53.3%	17.21	0.028
Welfare spending, GSS paper and pencil	11	1.048	0.013	1.038	0.037	0.0133	87.3%	85.19	0.000
Welfare spending, COVID-era experimental	10	0.434	0.018	0.433	0.021	0.0013	27.7%	12.39	0.192
Welfare spending, Huber benchmark	2	0.462	0.052	0.462	0.052	0.0000	0.0%	0.08	0.774
Welfare spending, Huber and GSS 1986	3	0.702	0.036	0.621	0.154	0.0669	94.1%	41.51	0.000
Welfare spending, COVID-era observational	2	0.362	0.030	0.362	0.030	0.0000	0.0%	0.73	0.392
Issue framing, pre-COVID	9	0.254	0.043	0.258	0.055	0.0106	39.0%	12.92	0.115
Issue framing, COVID era	27	0.094	0.015	0.094	0.015	0.0003	4.2%	36.04	0.091
Intentional side effects, pre-COVID	2	0.640	0.012	0.640	0.012	0.0000	0.0%	0.31	0.580

The pattern is the one the fixed-effect assumption invites. In 6 of the fifteen pools more than half the variance in the component estimates is between studies rather than within them, and where that is true the random-effects standard error is much wider, by up to a factor of 7.7. The point estimates move little. The reading is not that the article's benchmarks are wrong but that several of them pool studies run decades apart in different modes, and a single true effect across those is a strong assumption to make silently.

9 Figures

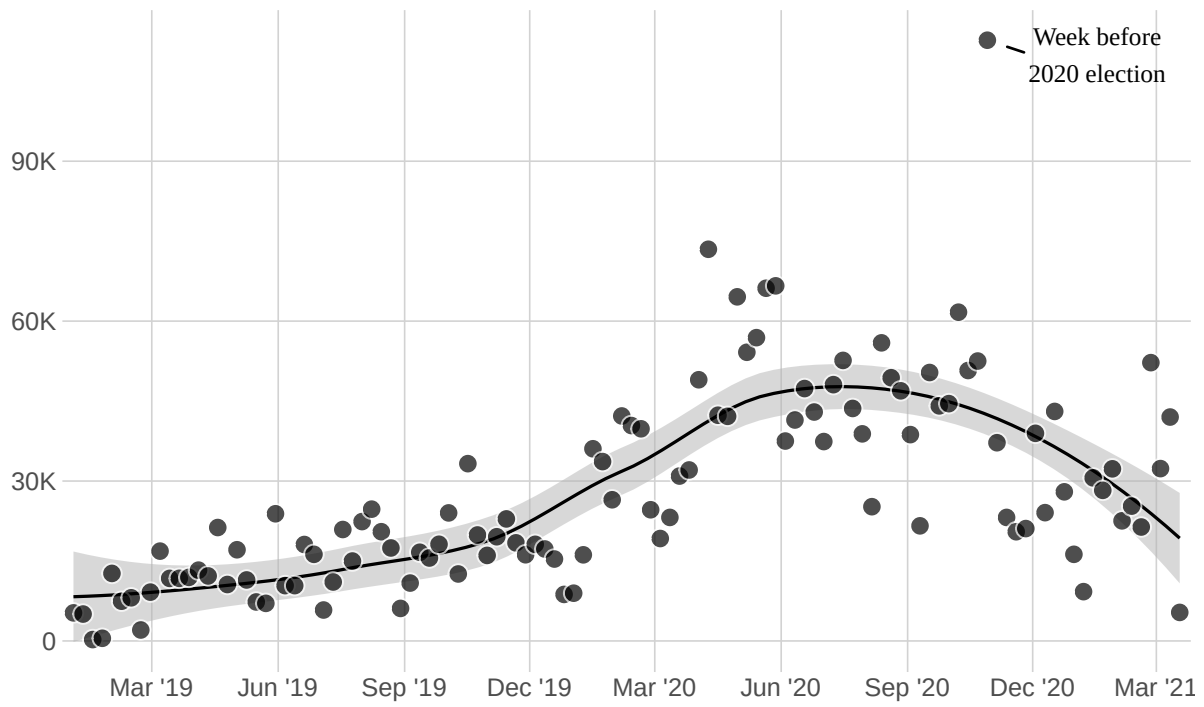


Figure 1: Figure 1: Total weekly Lucid survey responses to academic buyers, January 2019 to March 2021.

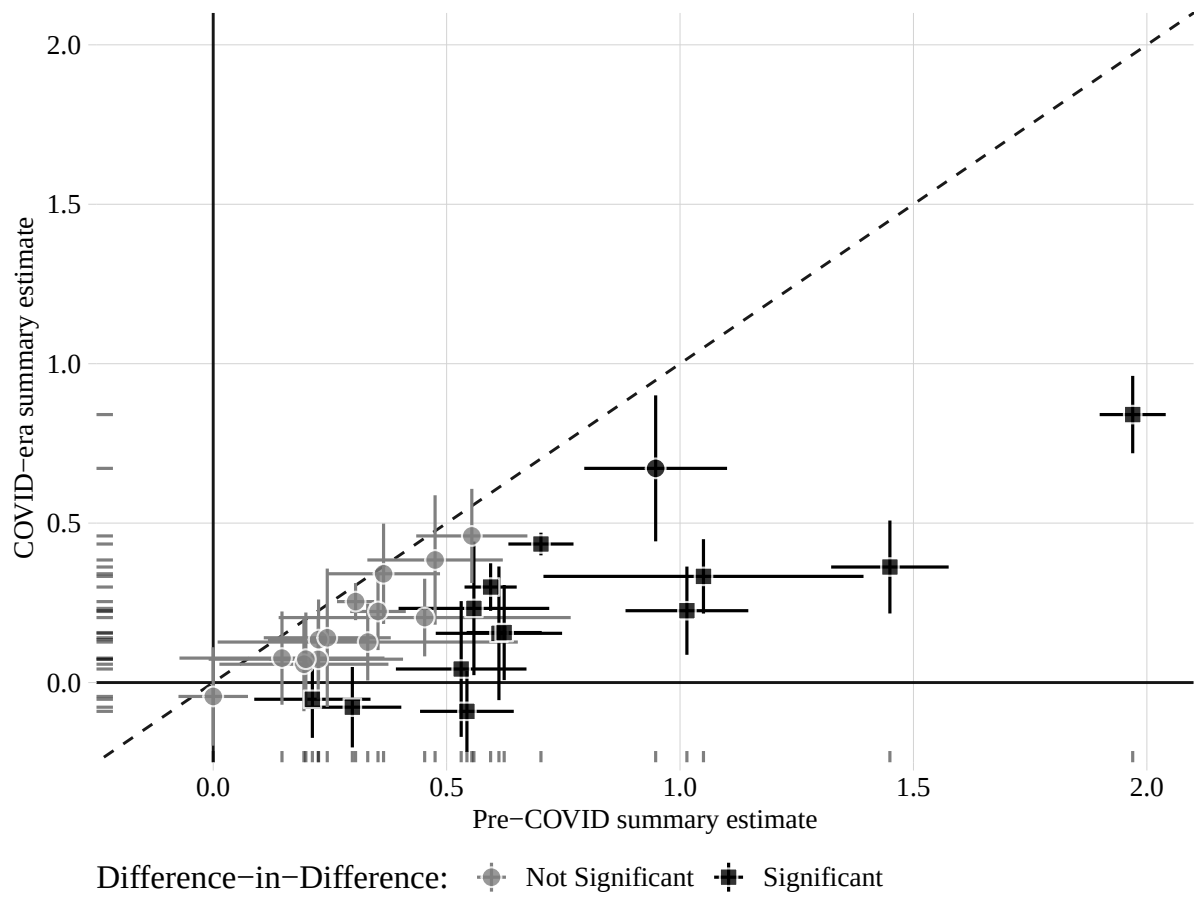


Figure 2: Figure 2: Comparison of 28 summary effect sizes across 11 studies, conjoint excluded. Fill marks a difference significant at $p < 0.05$; square points remain significant after controlling the false discovery rate.

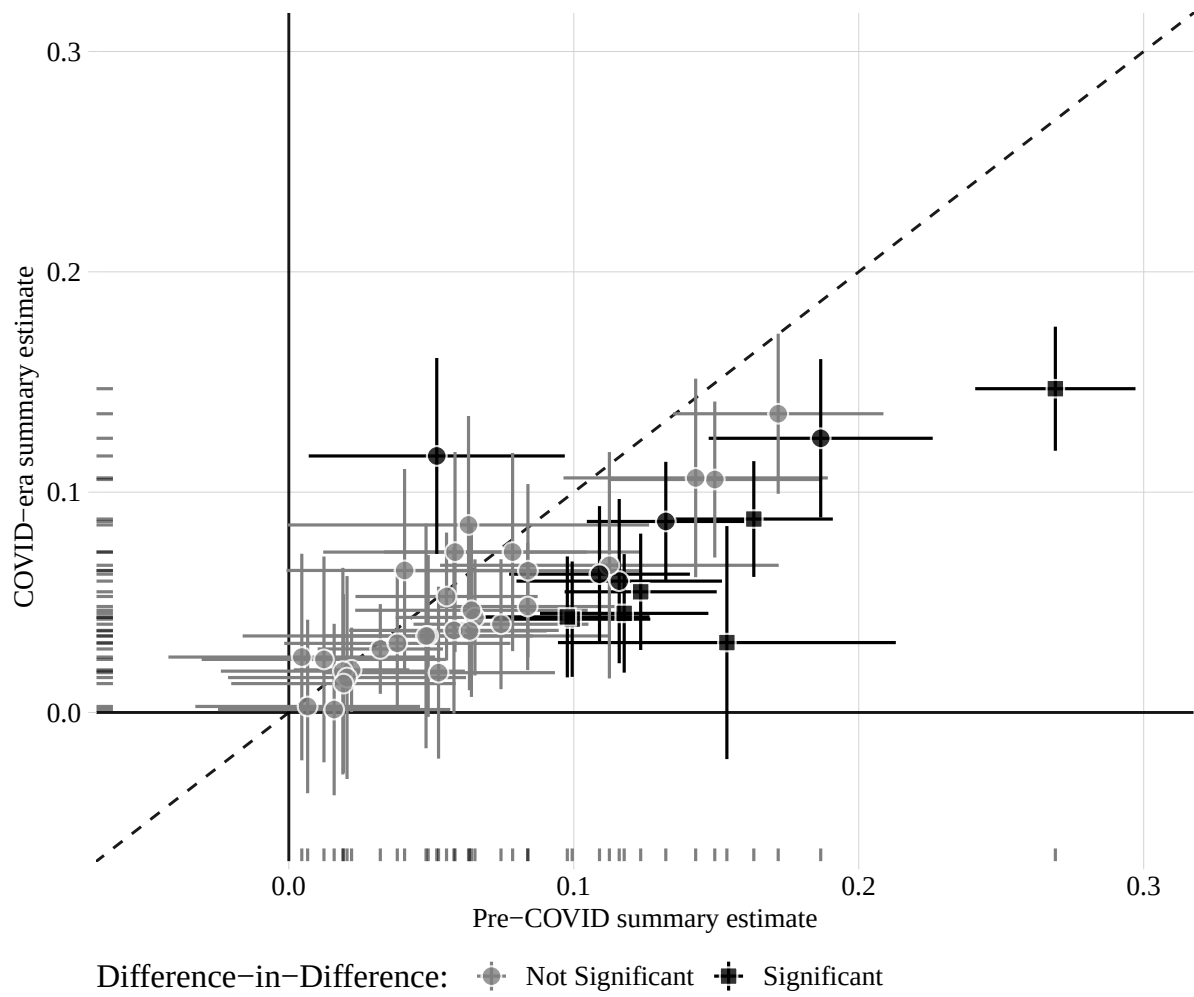


Figure 3: Comparison of 41 summary effect sizes in conjoint experiments, marked as in Figure 2.

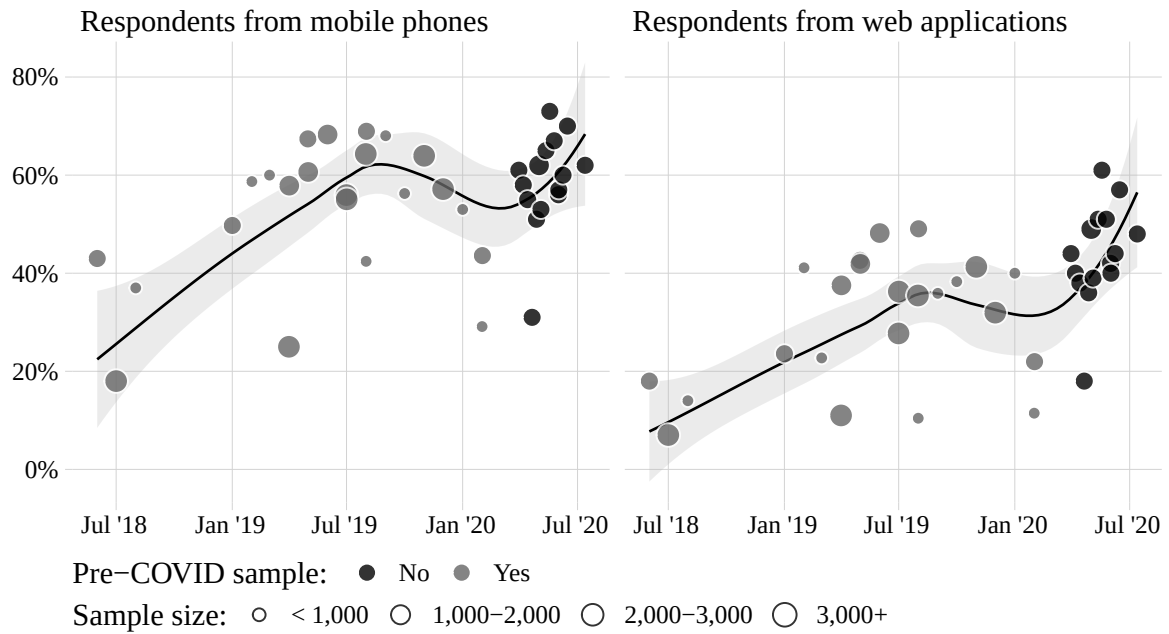


Figure 4: Figure 4: Respondents from mobile devices and web applications, June 2018 to July 2020.

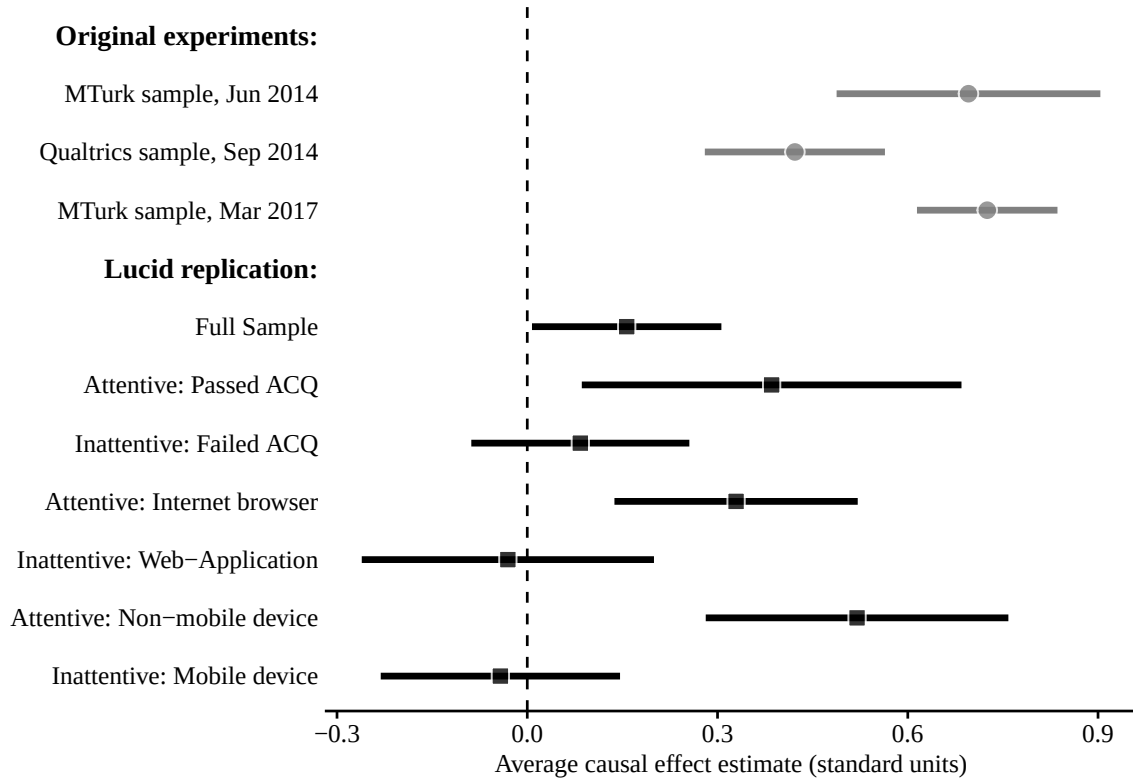


Figure 5: Figure 5: Reanalysis of estimated treatment effects on trust in government for the Peyton (2020) replication.

Figures 2 and 3 are the corrected versions. They encode significance before adjustment by fill and significance after the Benjamini-Hochberg adjustment by point shape, matching the figures printed in the corrigendum rather than those in the original article.

10 Table 2

Table 8: Attention-check pass rates by attentiveness and level of difficulty, reproducing published Table 2. Standard errors in parentheses; an asterisk marks a difference significant at $p < 0.05$.

Difficulty	Browser	Web app	Difference	Non-mobile	Mobile	Difference
Easy	0.66 (0.02)	0.54 (0.02)	0.11 (0.03)*	0.63 (0.02)	0.58 (0.02)	0.05 (0.03)
Medium	0.53 (0.02)	0.37 (0.02)	0.16 (0.03)*	0.50 (0.02)	0.42 (0.02)	0.08 (0.03)*

Hard	0.22 (0.02)	0.15 (0.01)	0.07 (0.02)*	0.24 (0.02)	0.16 (0.01)	0.08 (0.02)*
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11 Maintained rewrite verification

329 of the 358 claims the rewrite can compute match the published value at the page’s own precision, and 53 of the 59 claims about shape, sign or count hold. The 9 rows carrying no verdict are **approximate** hedges, which have no comparison by design, and the descriptive rows are counted separately above.

The pipeline is deterministic. Running `run_all.R` twice in succession returns every file it writes byte-identical, the figure PDFs included: `maintained/helpers.R` blanks the wall-clock `/CreationDate` and `/ModDate` that R’s `pdf()` device stamps into every figure, which are otherwise the only reason two runs differ.

The coverage gate was tested by breaking it three ways. Deleting a claim block fails the count. Changing a computed value in the claims file fails the cross-instrument comparison. Changing a recorded precision in the extraction fails the round-trip check, which runs before anything consumes the precision, so a wrong precision cannot hide behind a value comparison downstream.

12 R environment

Table 9: R version and key package versions at the last run.

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23
Date run	2026-08-10
tidyverse	2.0.0
estimatr	1.0.6
metafor	5.0.1
ggthemes	5.2.0
ggrepel	0.9.8
ggforce	0.5.0
lemon	0.5.2
RCurl	1.98.1.19
haven	2.5.5
xtable	1.8.8

knitr	1.51
kableExtra	1.4.0
here	1.0.2
